

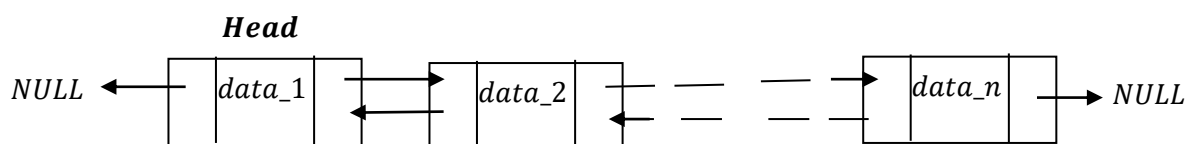
Doubly Linked List – Introduction

A doubly linked list is a linked list in which a node contains a pointer to the previous as well as the next node in the sequence. Therefore, a doubly linked list is also called two – way linked list. It consists of three parts : data, a pointer to the next node in sequence (next pointer), a pointer to the previous node(previous pointer). Each node has two links: one point to the previous node, or points to a NULL value or empty list if it is the first node, and one point to the next, or points to a NULL value or empty list if it is the final node.

The advantage of a doubly linked list is that given a node in the list, we can navigate in both directions. A node in a singly linked list cannot be removed unless we have the pointer to its predecessor. But in a doubly linked list, we can delete a node even if we don't have the previous node address (since each node has a left pointer pointing to the previous node and can remove backward).

The primary disadvantages of doubly linked lists are:

- → *Each node requires an extra pointer ,requiring more space.*
- → *The insertion or deletion of a node takes a bit longer(for more pointer operation).*



***A simple representation of Doubly Data
Linked List .***
